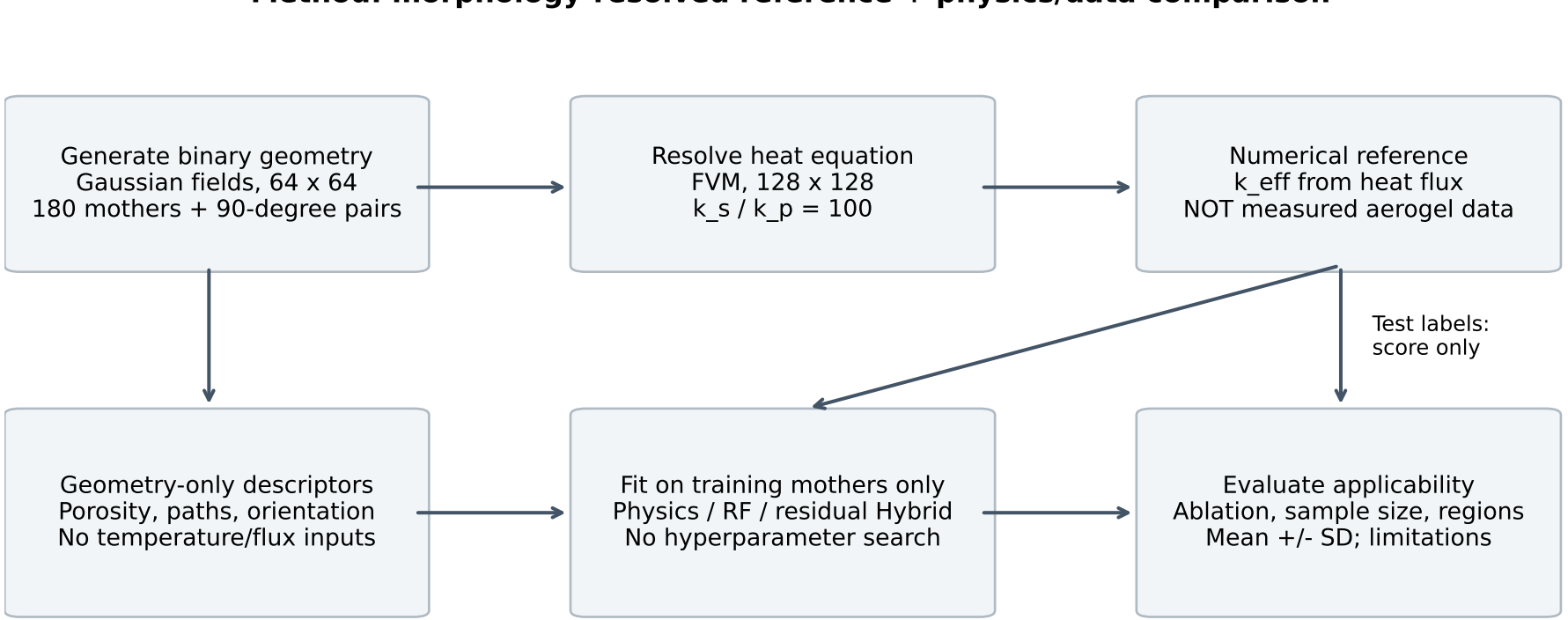
Method: morphology-resolved reference + physics/data comparison
Generate binary geometry
Gaussian fields, 64 x 64
180 mothers + 90-degree pairs
Resolve heat equation
FVM, 128 x 128
k_s / k_p = 100
Numerical reference
k_eff from heat flux
NOT measured aerogel data
Test labels:
score only
Geometry-only descriptors
Porosity, paths, orientation
No temperature/flux inputs
Fit on training mothers only
Physics / RF / residual Hybrid
No hyperparameter search
Evaluate applicability
Ablation, sample size, regions
Mean +/- SD; limitations